


# **L12a**: How Do Micro Decisions Lead to Emergent Macro Trends?

CHEME 5660 Cornell University

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# Objectives for Today

Today, we write down simple rules for individual traders, simulate many of them together, and measure what the market as a whole does.

## Today's concepts

- Agent-based models: the five parts every model shares, in the minority game and the Farmer–Joshi market.
- The Boltzmann choice rule: the maximum-entropy choice, with a temperature that sets how strongly agents favor high scores.
- Micro to macro: volatility in the minority game; fat tails and volatility clustering in the Farmer–Joshi market.

# What You Need to Know

## Core concepts

- Five parts: agents, information, a decision rule, an aggregate, and feedback from the aggregate to the next decisions.
- Maximum entropy for a fixed expected score gives $p_j = \exp(U_j/T)/\mathcal{Z}$: greedy as $T \to 0$, uniform as $T \to \infty$.
- The minority game's volatility per agent depends on $\alpha$ and $T$; Farmer–Joshi value investors and trend followers together produce fat tails and volatility clustering.

## For understanding (you won't reproduce these)

- The critical value $\alpha_c \approx 0.3374$ (cited, not derived).
- The minority-game price, an extension of the original model.
- The Farmer–Joshi threshold rules and our parameter values.

# Lectures and Examples

**Lecture:**

CHEME-5660-L12a-Lecture-AgentBasedMarkets-Fall-2026.ipynb

**Examples:**

CHEME-5660-L12a-Example-ThermalMinorityGame-Fall-2026.ipynb

CHEME-5660-L12a-Example-FarmerJoshi-Fall-2026.ipynb

# In the News: The November 5 Sell-Off

On Thursday, November 5, 2026, the Nasdaq fell 5.5% and the S&P 500 about 4.4%, the steepest one-day declines since April 2025, on worries that AI data-center spending could slow:

- The seven largest technology companies: lost more than 1 trillion USD of market value.
- Retail investors: sold a net 338 million USD of individual stocks, the largest one-day outflow since March 2020. “On a typical sell-off, retail buy the dip. Today, that bid has effectively disappeared from individual names.”

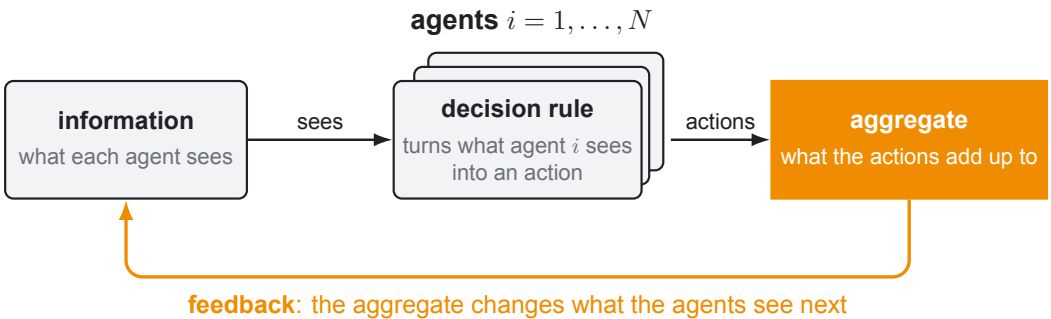
**Connection:** the day’s move was the sum of individual buy, sell, or wait decisions. On 2008: “the fault lies not in the stars, but in us.”

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**Source:** CNBC; CNBC, Vanda Research data; FCIC report

# Five Parts of Every Agent-Based Model

Agents see information, their decision rules turn it into actions, the actions add up to an aggregate, and the aggregate feeds back into what they see next:



# The Structure of an Agent-Based Model

An agent-based model describes individual decision makers and their rules, simulates them together, and measures the market-wide result:

| Part | Minority game | Farmer–Joshi market |
|---|---|---|
| Agents | $N$ agents, $J$ fixed policies each | Value investors, trend followers |
| Information | History $\mu_t$ of the last $m$ outcomes | Mispricing or price change over a lag |
| Decision rule | Choose a policy by its score | Enter or exit at a threshold |
| Aggregate | Attendance $A_t$ | Net order, then the price |
| Feedback | $A_t$ updates scores and history | New price changes every signal |

# The Minority Game

Each round, $N$ (odd) agents buy ($+1$) or sell ($-1$); the minority side, $-\text{sign}\, A_t$, wins. Agent $i$ takes the action $a_{i,t} = \pi_{ij}(\mu_t)$ of one of its $J$ policies, tables $\pi_{ij}(\mu)$ over the $P = 2^m$ histories $\mu$:

$$A_t = \sum_{i=1}^{N} a_{i,t} \qquad\qquad U_{ij,t+1} = U_{ij,t} - \frac{\pi_{ij}(\mu_t) A_t}{P}$$

where the attendance $A_t$ is the number of buyers minus sellers and $U_{ij,t}$ is the score of policy $\pi_{ij}$. A policy that would have put its agent in the minority gains score.

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**Source:** Challet and Zhang (1997); scores divided by $P$ as in Challet, Marsili, and Zecchina (2000)

# Volatility per Agent

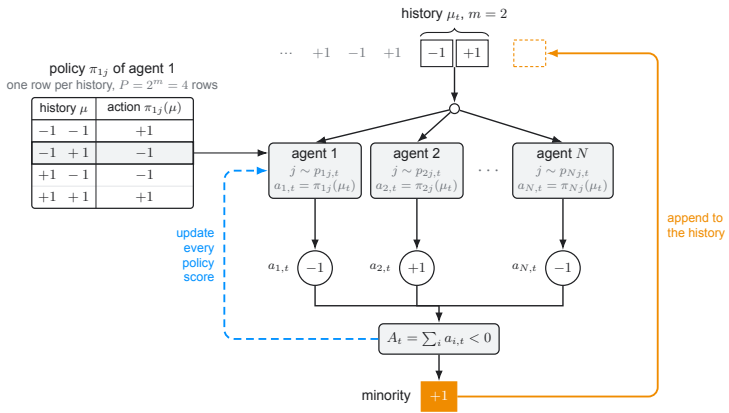
Each coin flip is a Bernoulli trial: $X_{i,t} = 1$ (buy) with probability $1/2$, and its action $a_{i,t} = 2X_{i,t} - 1$ has mean 0 and variance 1. The attendance adds the $N$ agents' independent actions, so it has variance $N$, the baseline for the volatility per agent:

$$\frac{\sigma_A^2}{N} = \frac{\langle A^2 \rangle}{N} \qquad\qquad \alpha = \frac{P}{N}$$

where $\langle \cdot \rangle$ averages over rounds and $\alpha$ is the number of histories per agent. $\sigma_A^2/N > 1$: more agents on the same side than with coin flips; $\sigma_A^2/N < 1$: a more even split. In the greedy game, each agent uses its best-scoring policy. When $\alpha$ is small, the game is crowded: many agents switch to the same action at the same time.

# One Round of the Minority Game

The history goes to every agent; the actions add up to the attendance, which feeds back into the history and every policy's score:



# Adding a Price: Instructor's Extension

The original game has no price. A market maker treats the attendance as the excess demand and moves the log price, as in the Farmer–Joshi market:

$$\ln S_{t+1} = \ln S_t + \frac{A_t}{\lambda} + \sigma_S \sqrt{\Delta t}\, Z_{t+1}$$

where $\lambda$ is the liquidity, $Z_{t+1} \sim \mathcal{N}(0,1)$ is independent noise, and $\sigma_S$ is its annualized volatility. If one round lasts $\Delta t$ years, the growth rate is $g_{t+1} = \frac{1}{\Delta t} \ln \frac{S_{t+1}}{S_t}$, with mean square about $\sigma_A^2/(\lambda^2 \Delta t^2) + \sigma_S^2/\Delta t$. At fixed $N$, $\lambda$, and $\sigma_S$, a crowded game moves the price more.

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**Related:** Challet, Chessa, Marsili, and Zhang (2000), with the opposite sign and no noise

# The Boltzmann Choice Rule

An agent chooses policy $\pi_j$ with probability $p_j$. The entropy $\mathcal{H}(p) = -\sum_j p_j \ln p_j$ measures how spread out the choice is: zero for a certain choice, $\ln J$ for a uniform one.

**Theorem:** among all $p_j$ with a given expected score $\sum_j p_j U_j = \bar{U}$, strictly between the average and the largest score, the one with the largest entropy is

$$\boxed{p_j = \frac{1}{\mathcal{Z}} \exp\left(\frac{U_j}{T}\right)}, \qquad \mathcal{Z} = \sum_{k=1}^{J} \exp\left(\frac{U_k}{T}\right)$$

where $T > 0$ is the temperature, in the units of the scores, and $\mathcal{Z}$ is the partition function.

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**Source:** Cavagna, Garrahan, Giardina, and Sherrington (1999); proof in the optional advanced notebook

# What the Temperature Does

The temperature is the value at which the Boltzmann probabilities give the target expected score $\bar{U}$. Lowering $T$ raises $\bar{U}$:

- $T \to 0$: the best policy has probability 1, the greedy minority game; $\bar{U}$ approaches the largest score.
- $T \to \infty$: every policy has probability $1/J$; $\bar{U}$ is the average score.
- In between: with two scores $\Delta$ apart, the better policy is used with probability $e^{\Delta/T}/(e^{\Delta/T}+1)$, about 0.73 at $T = \Delta$.

# The Thermal Minority Game Algorithm

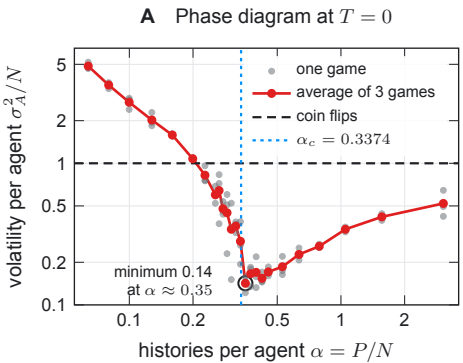
**Initialize:** $N$ (odd), $m$, $J$, $T$, rounds $R$, $\lambda$, $\sigma_S$, $\Delta t$; $P = 2^m$; random tables $\pi_{ij}(\mu)$; scores $U_{ij,0} \leftarrow 0$; random $\mu_0$; starting price $S_0$. For each round $t = 0, \dots, R-1$ **do**:

1. Choose: draw $j_{i,t}$ with $p_{ij,t} = \exp(U_{ij,t}/T)/\mathcal{Z}_{i,t}$.
2. Act: $a_{i,t} \leftarrow \pi_{i\, j_{i,t}}(\mu_t)$.
3. Aggregate: $A_t \leftarrow \sum_i a_{i,t}$; the minority action is $-\operatorname{sign} A_t$.
4. Score: $U_{ij,t+1} \leftarrow U_{ij,t} - \pi_{ij}(\mu_t) A_t / P$ for every $i$ and $j$.
5. History: drop the oldest action in $\mu_t$, append $-\operatorname{sign} A_t$.
6. Price (extension): draw $Z_{t+1} \sim \mathcal{N}(0,1)$; $\ln S_{t+1} \leftarrow \ln S_t + A_t/\lambda + \sigma_S \sqrt{\Delta t}\, Z_{t+1}$.

**Output:** $A_0, \dots, A_{R-1}$, $S_0, \dots, S_R$, and $\sigma_A^2/N = \langle A^2 \rangle / N$ after a burn-in (the first rounds, discarded).

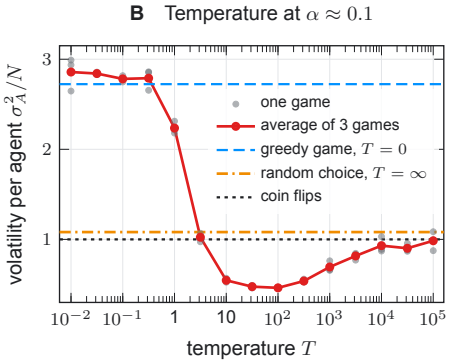
# The Phase Diagram

Volatility per agent against $\alpha$ in the greedy game ($T = 0$). The minimum, 0.14 at $\alpha \approx 0.35$, is close to $\alpha_c \approx 0.3374$:



# The Effect of Temperature

At $\alpha \approx 0.1$: greedy 2.72, random choice 1.08, and 0.46 at $T = 100$, below both limits. Our scores are divided by $P$, so our $T$ is $1/\alpha$ times that of Cavagna and coauthors:



# The Farmer–Joshi Market

Trader $i$ holds $x_t^{(i)} \in \{-c_i, 0, +c_i\}$ and orders $\omega_{t+1}^{(i)} = x_{t+1}^{(i)} - x_t^{(i)}$. The market maker moves the log price in proportion to the net order:

$$\ln S_{t+1} = \ln S_t + \frac{1}{\lambda} \sum_i \omega_{t+1}^{(i)} + \sigma_S \sqrt{\Delta t}\, Z_{t+1}$$

where $\lambda$ is the liquidity, $Z_{t+1} \sim \mathcal{N}(0,1)$ is independent noise, and $\sigma_S$ is its annualized volatility. As in the minority game, the growth rate is $g_{t+1} = \frac{1}{\Delta t} \ln \frac{S_{t+1}}{S_t}$.

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**Source:** Farmer and Joshi (2002)

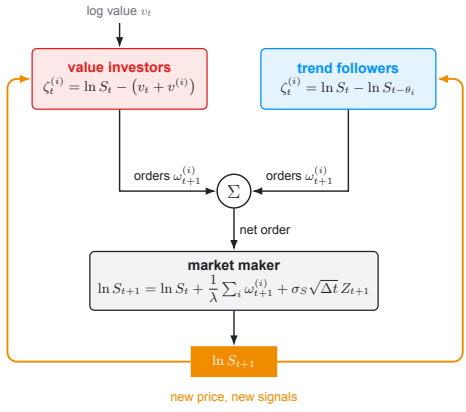
# Value Investors and Trend Followers

Each trader enters at a signal beyond $\pm\chi_i^{\text{in}}$ and exits back across $\pm\chi_i^{\text{out}}$, with position size $c_i = a(\chi_i^{\text{in}} - \chi_i^{\text{out}})$:

- Value investors: signal $\ln S_t - (v_t + v^{(i)})$; sell when overpriced, buy when underpriced.
- Trend followers: signal $\ln S_t - \ln S_{t-\theta_i}$; buy after a rise, sell after a fall.

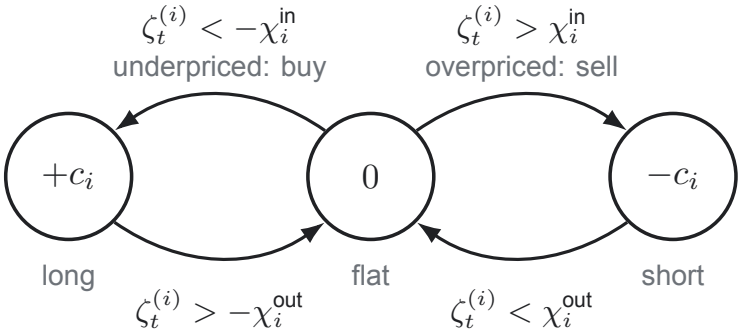
# One Step of the Market

Signals set positions, positions set orders, and the net order sets the price, which changes every signal at the next step:



# The Threshold Rule

A value investor's position is one of three states. It enters when the mispricing passes $\pm\chi_i^{\text{in}}$ and exits when it comes back past $\pm\chi_i^{\text{out}}$:



# The Farmer–Joshi Algorithm

**Initialize:** thresholds $\chi_i^{\text{in}} > \chi_i^{\text{out}}$, offsets $v^{(i)}$, lags $\theta_i$, $a$, $\lambda$, $\sigma_S$, $\sigma_v$, $\Delta t$, steps $R$;
$c_i \leftarrow a(\chi_i^{\text{in}} - \chi_i^{\text{out}})$; $x_0^{(i)} \leftarrow 0$. For each step $t = 0, \dots, R-1$ **do**:

1. Signals: $\zeta_t^{(i)} \leftarrow \ln S_t - (v_t + v^{(i)})$ (value) or $\ln S_t - \ln S_{t-\theta_i}$ (trend, once $t \geq \theta_i$).
2. Positions: the threshold rule sets $x_{t+1}^{(i)} \in \{-c_i, 0, +c_i\}$.
3. Orders: $\omega_{t+1}^{(i)} \leftarrow x_{t+1}^{(i)} - x_t^{(i)}$.
4. Price: draw $Z_{t+1} \sim \mathcal{N}(0,1)$; $\ln S_{t+1} \leftarrow \ln S_t + \frac{1}{\lambda}\sum_i \omega_{t+1}^{(i)} + \sigma_S\sqrt{\Delta t}\, Z_{t+1}$.
5. Value: draw $Z'_{t+1} \sim \mathcal{N}(0,1)$; $v_{t+1} \leftarrow v_t + \sigma_v\sqrt{\Delta t}\, Z'_{t+1}$.

**Output:** log prices, log values, growth rates $g_{t+1} = \frac{1}{\Delta t}\ln\frac{S_{t+1}}{S_t}$, volume.

# Stylized Facts: Simulated and Real

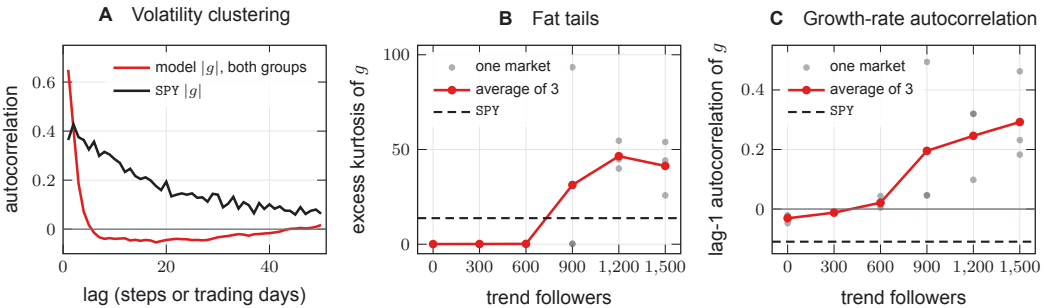
Both groups together, against `SPY` daily growth rates (kurtosis and autocorrelations do not depend on units):

| Statistic | Simulated | `SPY` |
|---|---|---|
| Excess kurtosis | 38.8 | 13.8 |
| Autocorrelation of $g$, lag 1 | 0.31 | $-0.11$ |
| Autocorrelation of $\|g\|$, lag 1 | 0.65 | 0.36 |
| Autocorrelation of $\|g\|$, lag 5 | 0.02 | 0.33 |
| Autocorrelation of volume, lag 1 | 0.55 | 0.75 |

Fat tails and clustering come out of the rules, but the clustering fades within a few steps.

# Clustering and the Mix of Traders

The model's clustering fades within a few steps (A). Fat tails and a positive growth-rate autocorrelation appear only with enough trend followers (B, C); every mix misses SPY by a wide margin on at least one statistic:



# Optional Advanced Material

**Maximum-entropy choice:** advanced/max_entropy/
CHEME-5660-L12a-Derivation-MaxEntropyChoice-Fall-2026.ipynb

We prove the Boltzmann choice rule with Lagrange multipliers and relate the temperature to thermodynamics. A derivation with no code.

**Biased policies:** advanced/biased_policies/
CHEME-5660-L12a-Derivation-BiasedPolicies-Fall-2026.ipynb

What if the policies lean toward buying? We derive how much of the bias agents who follow their scores can remove, and show that above a threshold the market locks into a steady trend. A derivation with no code.

# Summary

We introduced agent-based models, used the Boltzmann choice rule, the maximum-entropy choice, and connected trading rules to market-wide behavior.

## Key takeaways

- Agent-based models: agents, information, a decision rule, an aggregate, and feedback.
- Maximum entropy: $p_j \propto \exp(U_j/T)$ has the largest entropy for a given expected score; $T$ moves the agent from greedy to random.
- Micro to macro: the minority game's volatility depends on $\alpha$ and $T$, and a crowded game moves the price more; Farmer–Joshi traders produce fat tails and clustering, but not every stylized fact at once.